

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 150
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.1
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 150
samples_per_sec: 6
forecast_horizon_sec: 2
forecast_steps: 12
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-10 17:15:49

Training Start: 2025-05-10 17-02-09

Training End: 2025-05-10 17-09-13

Training Duration: 7.06 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 14976.1137

MSE: 383228895.5093

RMSE: 19576.2329

R2 Score: 0.7892

Multi-Step Forecast:

Forecast Steps: 12

MAE: 0.0197

MSE: 0.0006

RMSE: 0.0253

R2 Score: 0.9293

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 6,849 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

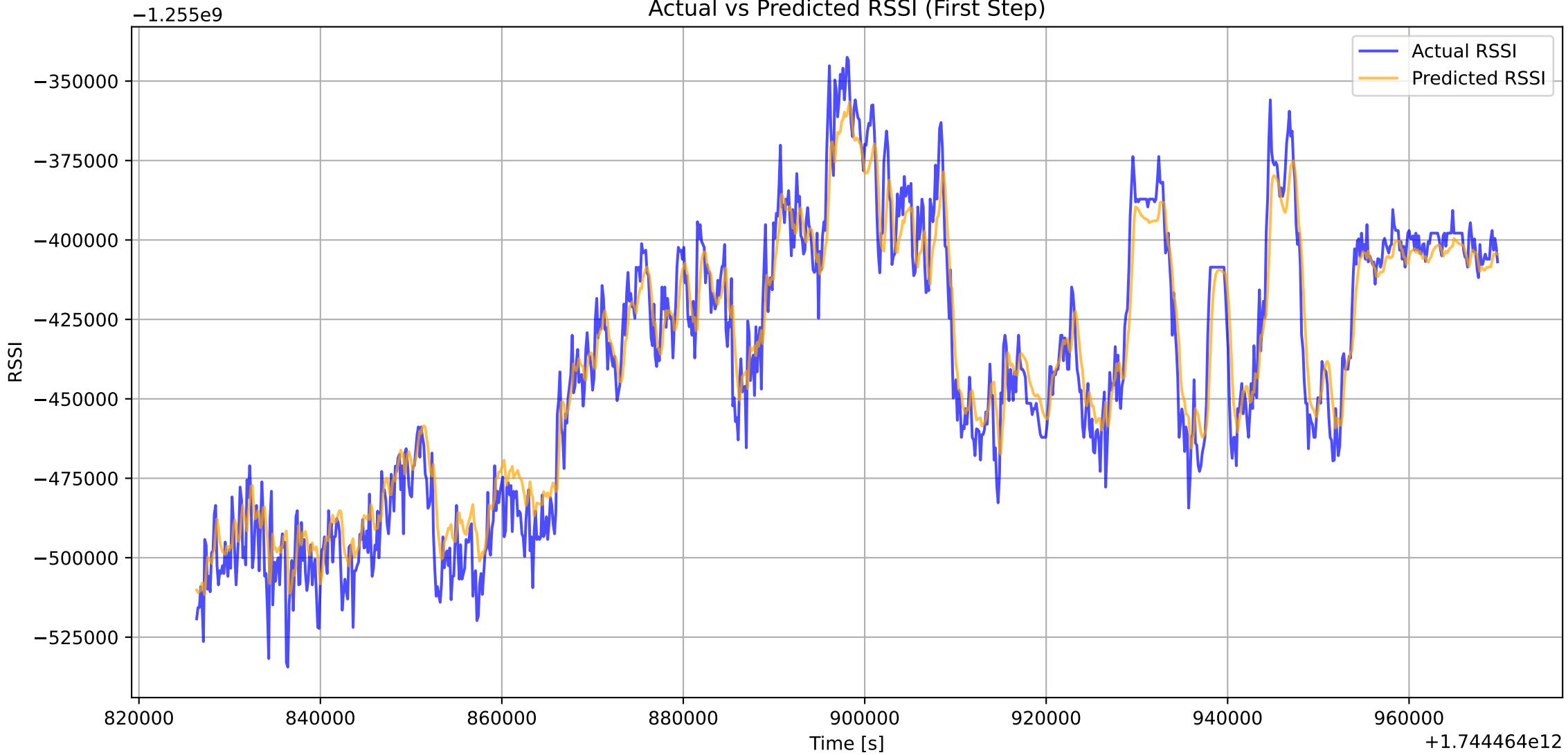
1744397085760
1744397086014
1744397086050
1744397086208
1744397086362

...

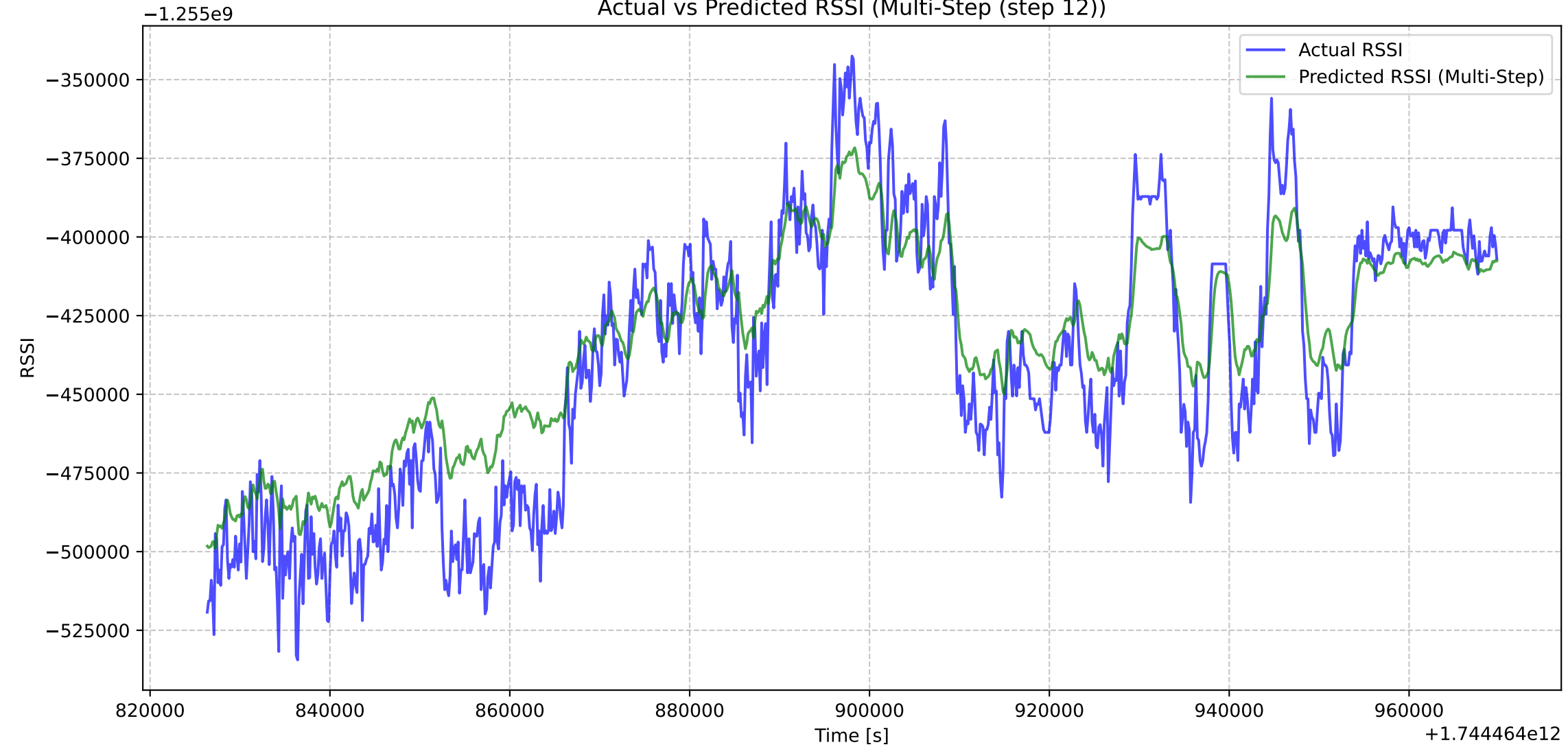
Last 5 timestamps AFTER:

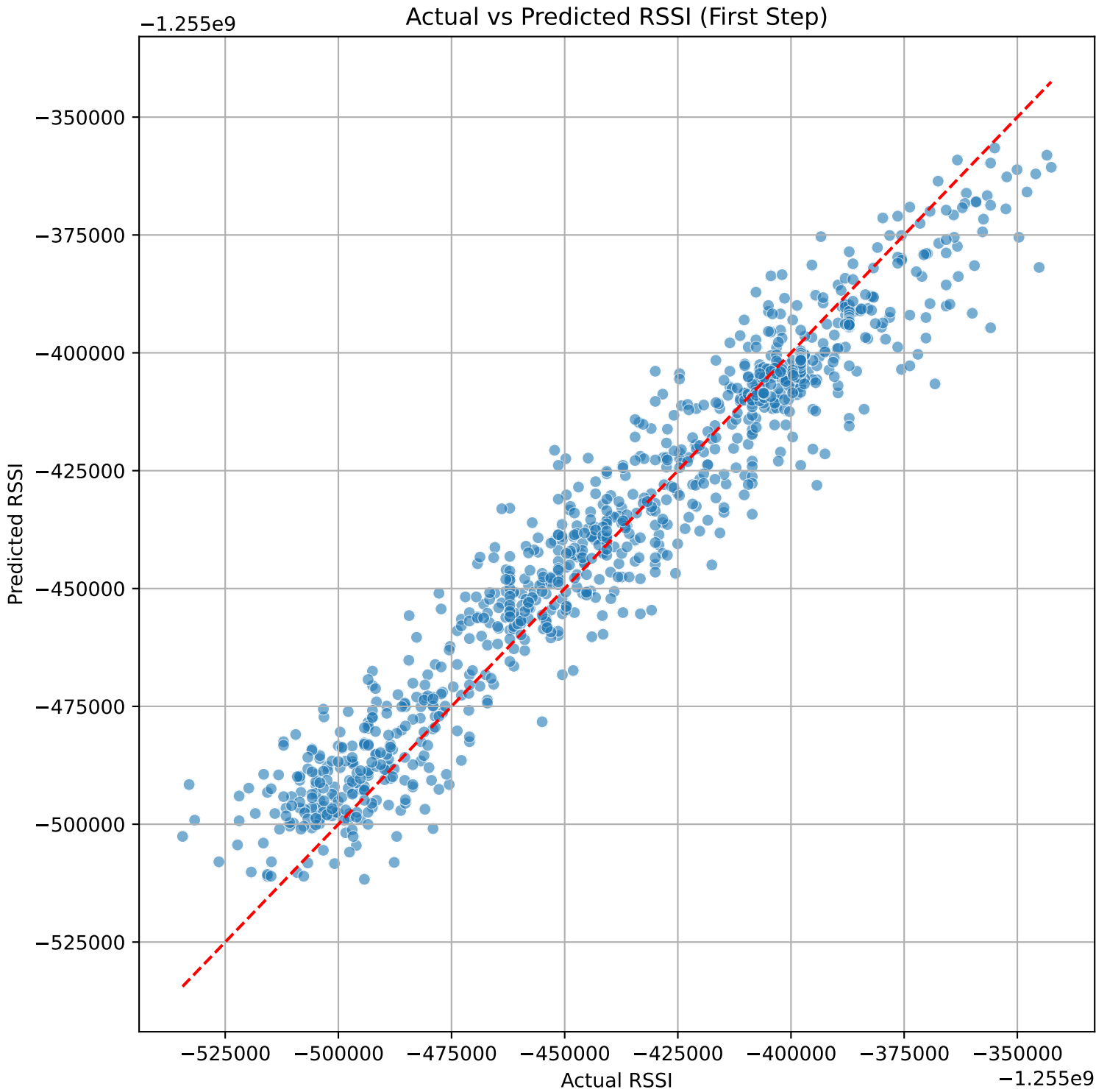
1744464970802
1744464970956
1744464971101
1744464971257
1744464971402

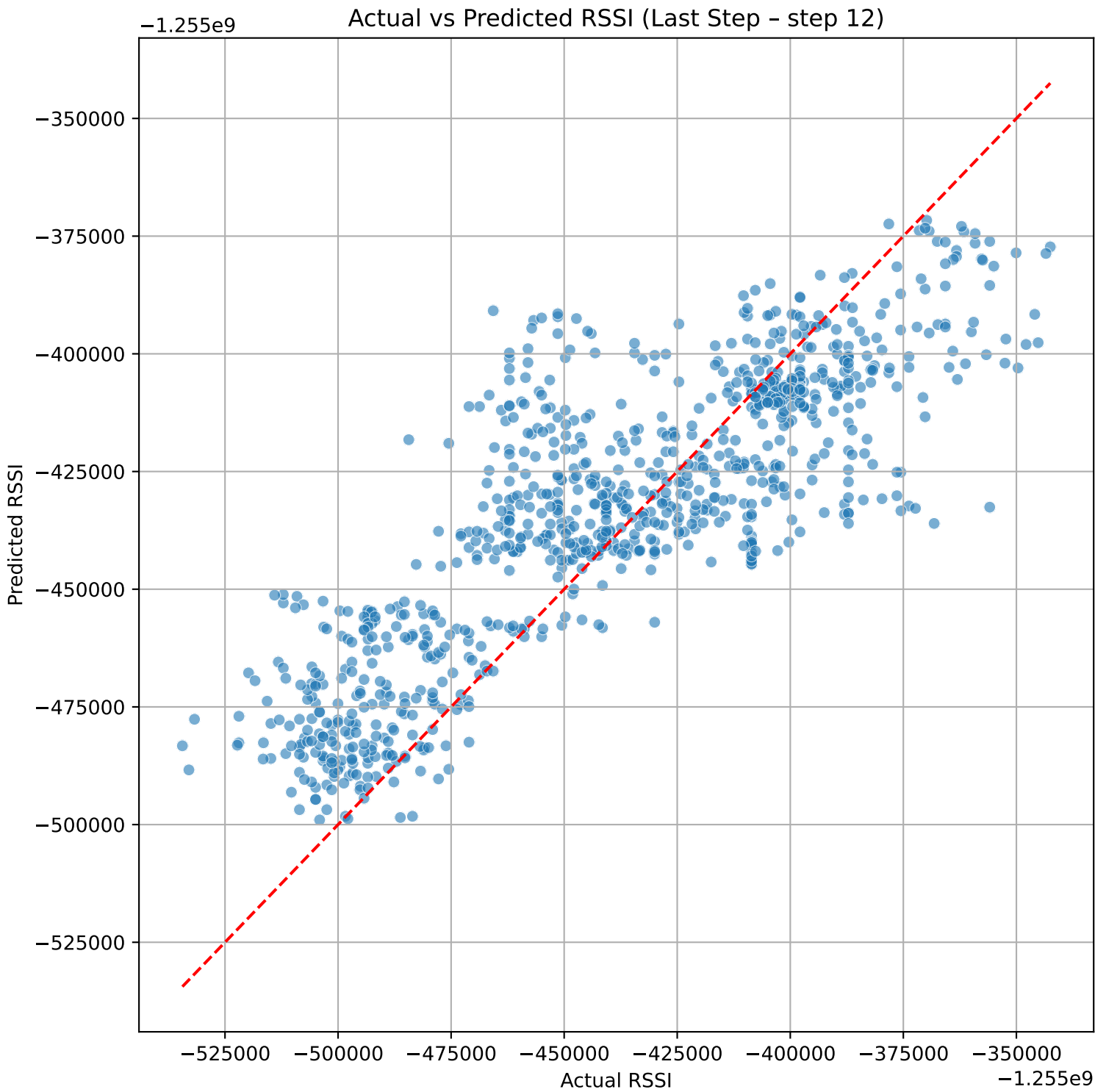
Actual vs Predicted RSSI (First Step)



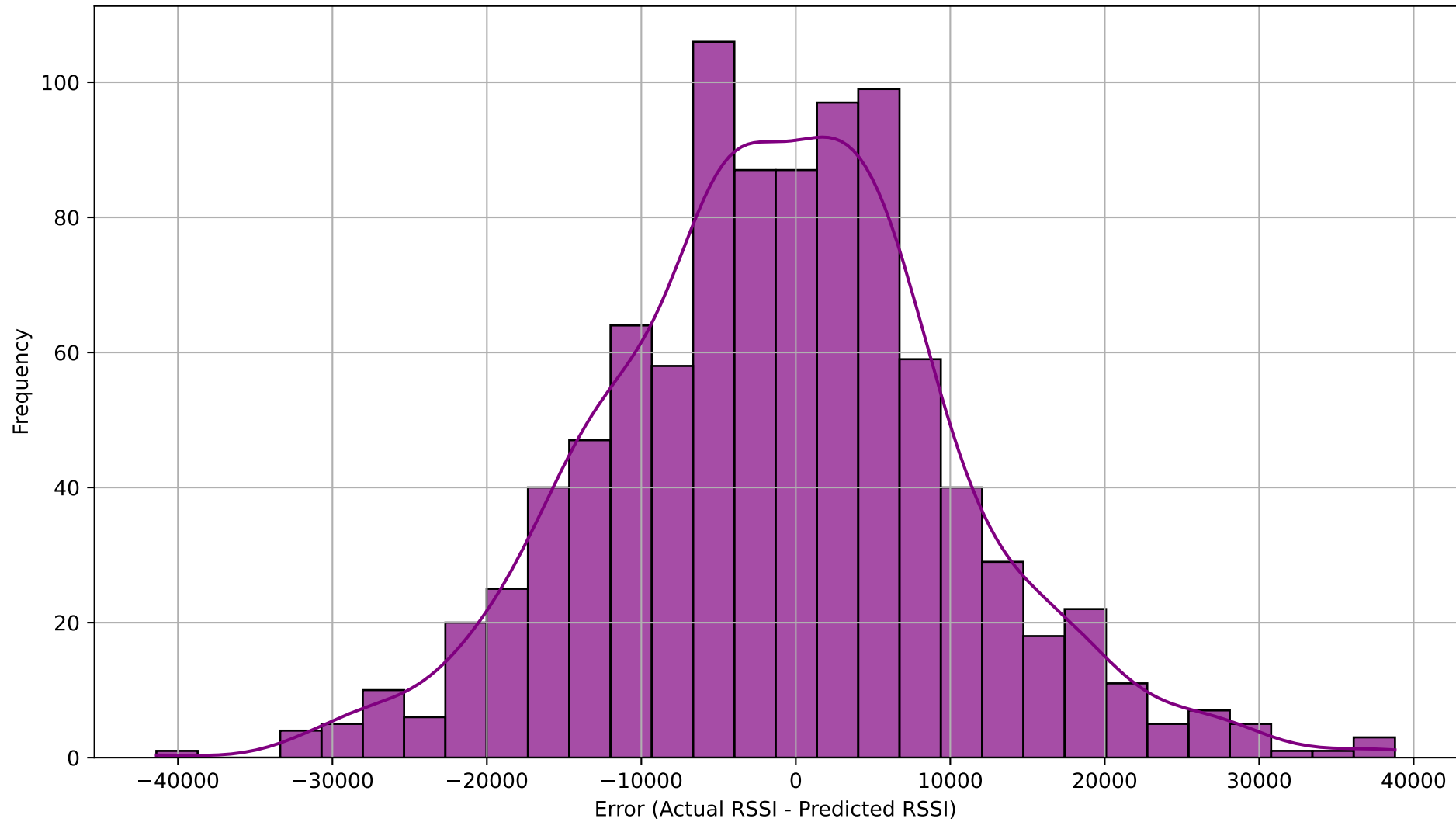
Actual vs Predicted RSSI (Multi-Step (step 12))



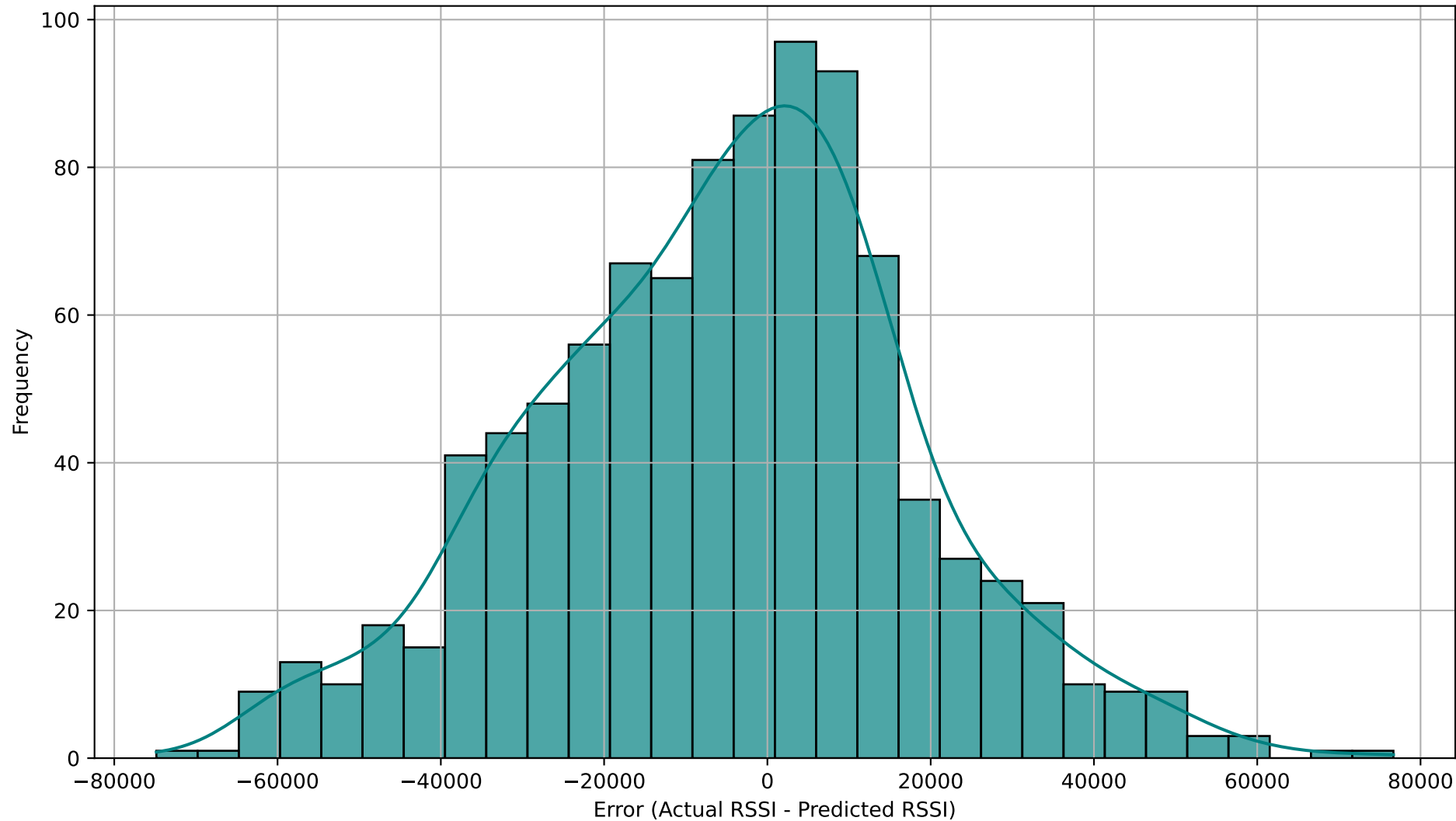




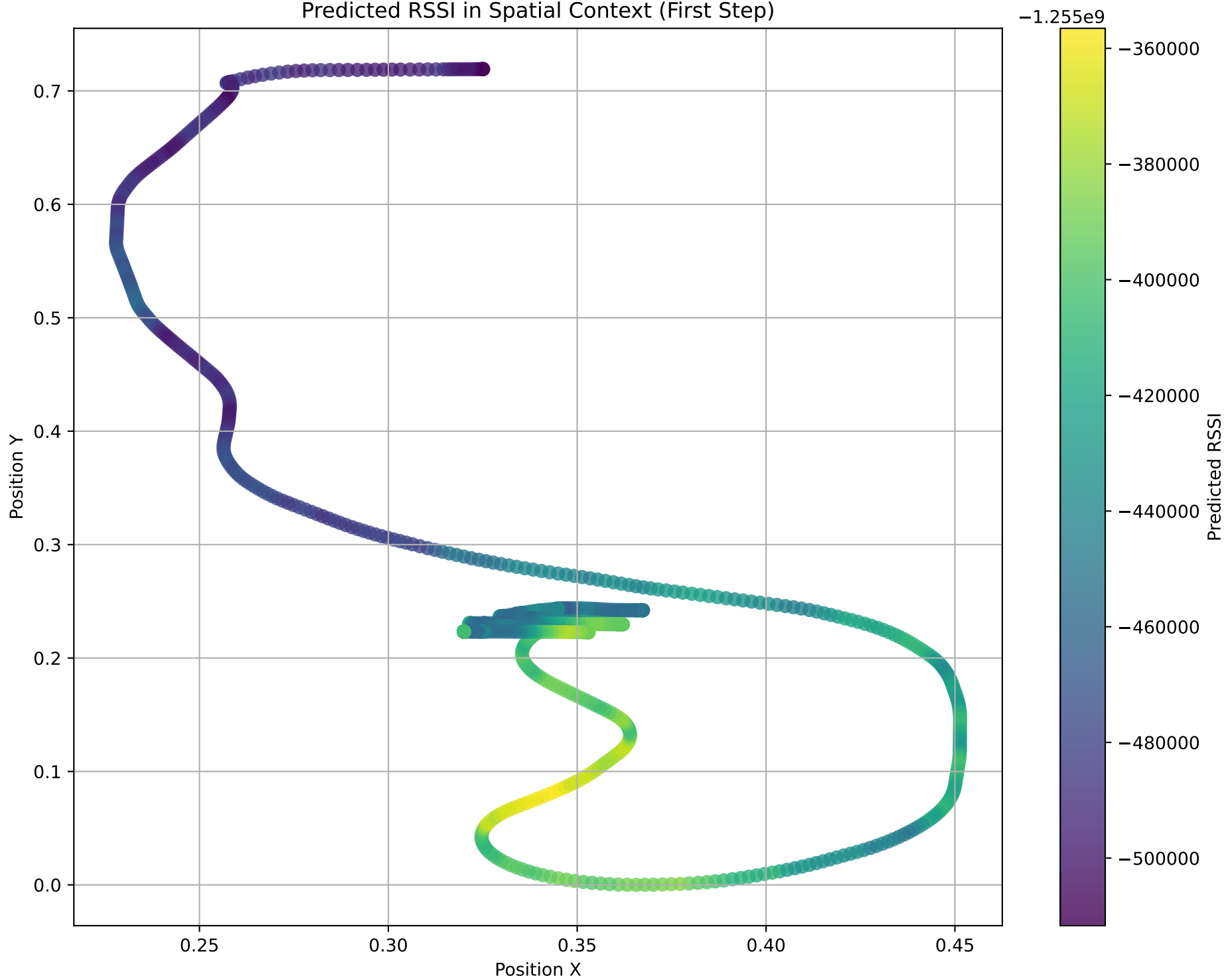
Distribution of Prediction Errors (First Step)



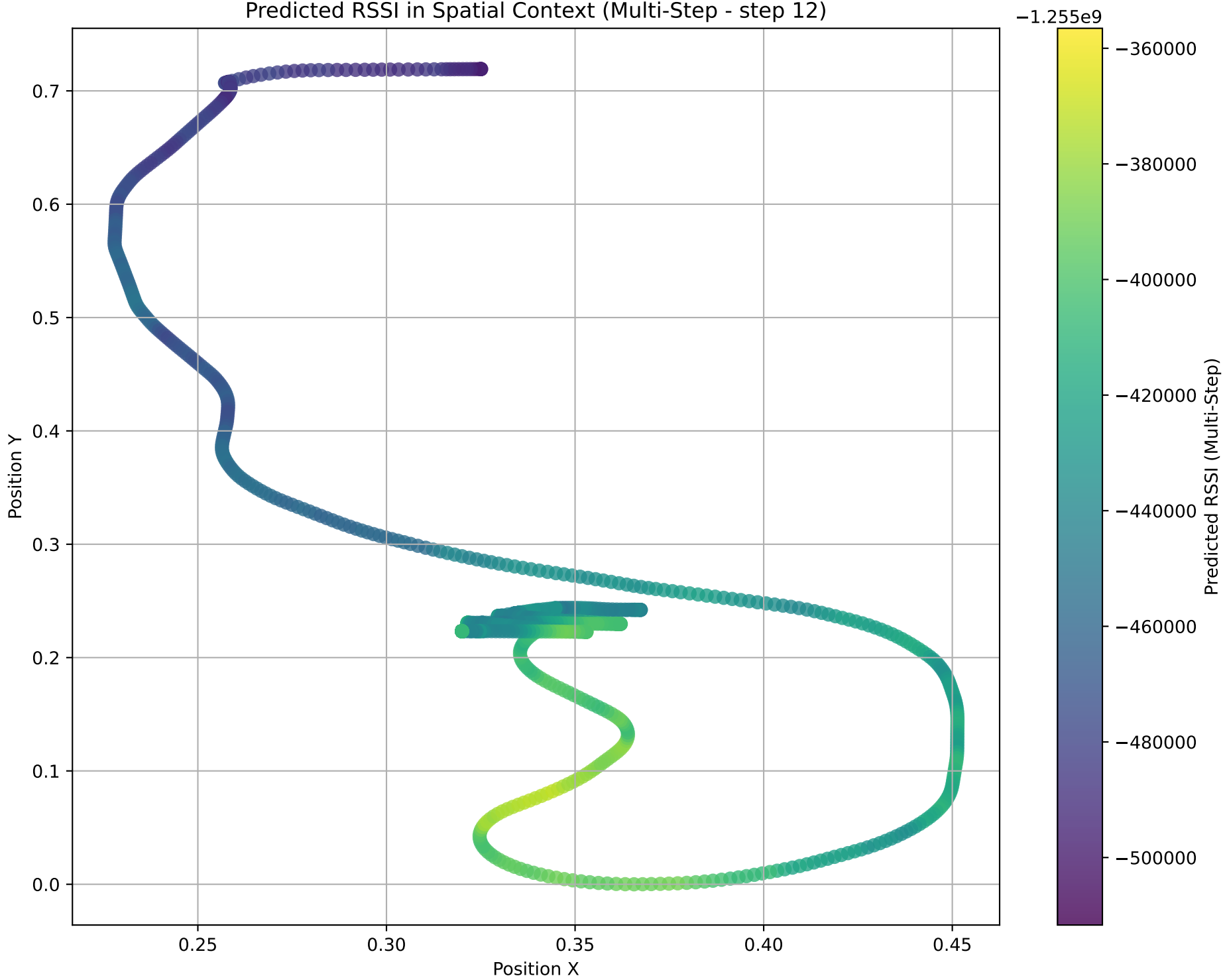
Distribution of Prediction Errors (Last Step - step 12)



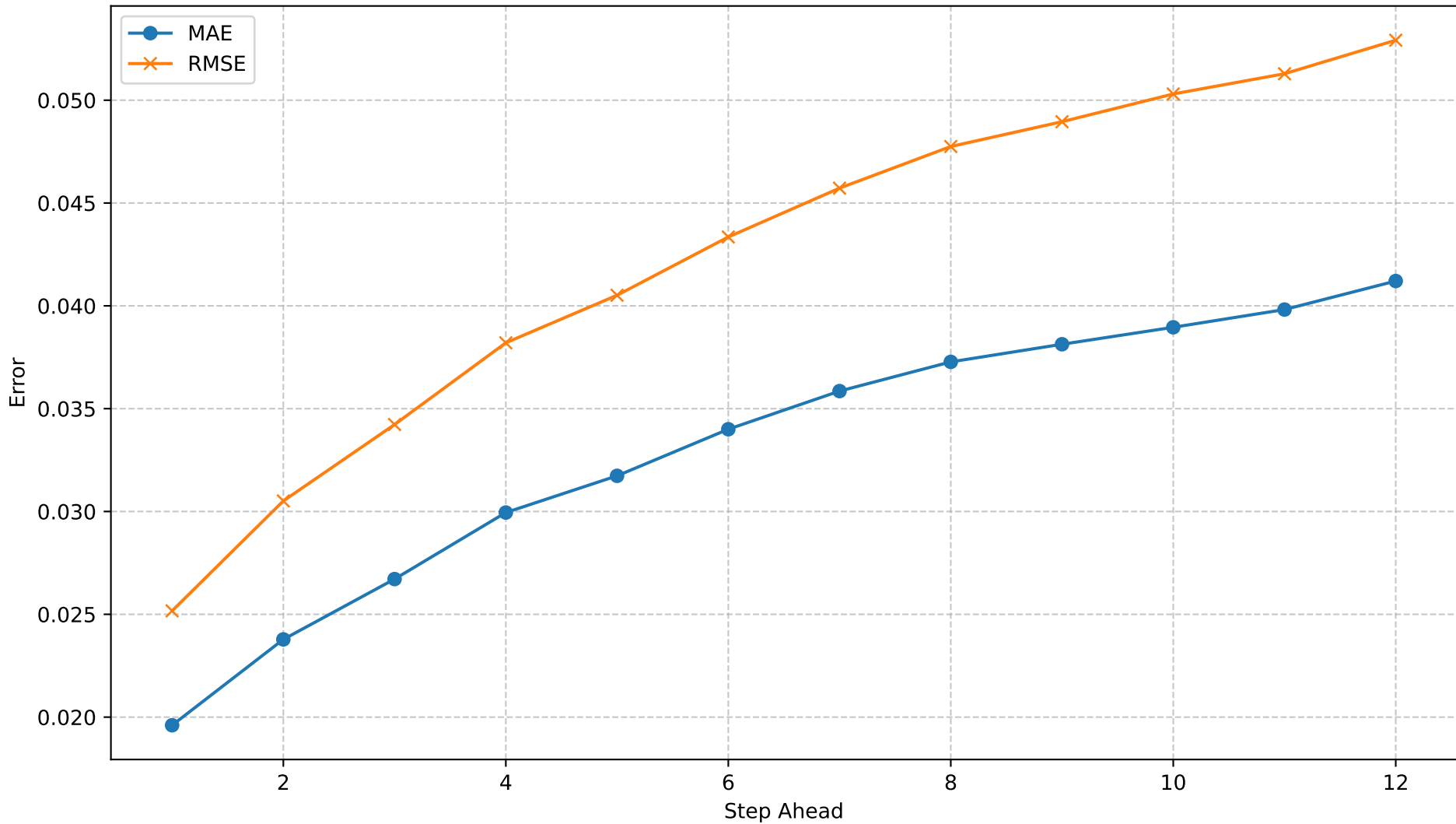
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 12)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 12

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.03
- Mediana : 0.03 / 0.04
- 12º passo : 0.04 / 0.05

Crescimento do erro: 110.2% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 12 (0.04), 11 (0.04), 10 (0.04)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.